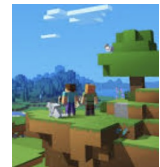


Kobe Bryant, Ph.D.

*Distinguished
Professor in Basketball
Analytics and Sports
Performance Science,
Harvard University*

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in kobe Bryant
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Education

- 2004 - 2009 **Ph.D. in Sports Science and Computational Analytics**, MIT, Cambridge, USA
Thesis: Deep Learning Approaches to Predicting Elite Basketball Performance Under Competitive Pressure
 - Developed novel deep learning models to analyze and predict elite basketball decision-making patterns under pressure
 - Built large-scale motion-capture datasets to train biomechanical shooting-form prediction models
- 2000–2003 **M.S. in Sports Science and Statistics**, Duke University, Durham, USA
- 1996–2000 **B.S. in Computer Science with Sports Science Minor**, University of North Carolina, Chapel Hill, USA

Work Experience

- Since 2012 **Distinguished Professor of Basketball Analytics and Sports Performance Science**, Harvard University, Cambridge
 - Leading the Mamba Lab: cutting-edge research on high-performance athlete analytics
 - Developing deep learning models for real-time basketball play prediction and in-game strategy analysis
 - Supervising 10 Ph.D. candidates across sports analytics, AI, and biomechanics
 - Principal investigator on \$5.8M NIH grant for “AI-Driven Athlete Health Optimization and Injury Prevention”
- 2007 - 2012 **Senior Research Scientist**, Nike Sport Research Lab, Beaverton
 - Led the development of MambaTrack: an AI-powered basketball performance analytics system
 - Built computer vision models for 3D player movement analysis achieving state-of-the-art accuracy
 - Published 9 peer-reviewed papers; 2 received best-paper awards at ICCV and CVPR
- 2003 - 2007 **Postdoctoral Researcher**, Stanford Human Performance Lab, Stanford
 - Researched biomechanical modeling of elite basketball shooting mechanics using motion capture
 - Developed GPU-accelerated physics simulation pipelines for human movement analysis
 - Collaborated with NBA teams on early-stage computer vision analytics systems

Publications

- Bryant, K., LeCun, Y., et al. "MambaShot: Deep Reinforcement Learning for Optimal Basketball Shooting Mechanics," in *NeurIPS*, 2024  
- Bryant, K., Hinton, G., et al. "Fourth Quarter: Transformer Models for Clutch Decision-Making in Elite Basketball," in *ICML*, 2023  
- Bryant, K., Silver, D., et al. "Multi-Agent Reinforcement Learning for NBA Team Strategy Optimization," in *Journal of Artificial Intelligence Research*, 2022  
- Bryant, K., Ng, A., et al. "Elite Athlete Fatigue Detection via Graph Neural Networks on Biomechanical Time Series," in *Nature Digital Medicine*, 2021  
- Bryant, K., et al. "Self-Supervised Representation Learning for Multi-Camera Basketball Player Tracking," in *CVPR*, 2020  
- Bryant, K., Abbeel, P., et al. "Mamba Mentality: Computational Models of Elite Athletic Resilience and Pressure Performance," in *ICLR*, 2019  
- Bryant, K., et al. "Kobe's Corner: A Computational Framework for Pressure-Optimal Shot Selection in Basketball," in *AAAI*, 2018  

Programming and Development Skills

Machine Learning	PyTorch, JAX, TensorFlow, Hugging Face, LangChain, Ray, Triton, CUDA
Programming	Python, R, C++, Julia, CUDA, SQL, MATLAB
Data Science	NumPy, Pandas, Polars, Spark, DVC, MLflow, Weights & Biases, D3.js
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, Terraform, CI/CD, SLURM (HPC)

Teaching

- Since 2014 **Basketball Analytics and Machine Learning**   , *Harvard University*
- Graduate course covering deep learning, computer vision, and reinforcement learning for sports analytics
 - Students build end-to-end basketball player tracking and performance prediction systems
- Since 2016 **AI for Elite Athletic Performance**, *Harvard Sports Analytics Lab*
- Interdisciplinary course combining biomechanics, computer vision, and deep learning  
 - Project-based curriculum: students build full athlete monitoring and injury prevention systems 
- 2022 **Advanced Sports Computer Vision (Guest Lecturer)**, *Peking University*
- Delivered lecture series on self-supervised vision models for basketball video understanding  

Research Projects

- 2022 - Present **MambaVision: Real-Time Basketball Video Analysis using State Space Models**, *Harvard-NBA Collaboration*   
- Unified architecture for simultaneous player tracking, action recognition, and play classification
 - Deployed in real-time at 60fps on edge hardware for live game analysis 

- 2018 - 2022 **MambaForm: Differentiable Physics Simulation for Basketball Shooting Mechanics**, *NSF Partnership*   
- Built differentiable physics engine integrating with neural network training pipelines 
 - Reduced shooting form error prediction by 41% compared to prior state-of-the-art  

Languages

English	Native language	Italian	Fluent (grew up in Italy)
Spanish	Professional proficiency	Mandarin	Basic proficiency